



# From glow to risk: degree of exposure of Arctiinae moths (Lepidoptera: Erebidae) to light pollution in the Neotropics

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## Abstract

The accelerated decline of insects is driven by anthropogenic pressures such as habitat loss, pesticide use, and climate change. Species vulnerability varies according to their exposure, sensitivity, and adaptive capacity to these threats. Artificial light at night (ALAN) further increases this vulnerability by disrupting nocturnal and crepuscular species, particularly moths. The subfamily Arctiinae is ecologically diverse, exhibits chemical defenses and ultrasonic signaling, and is considered a valuable bioindicator group. Here, we investigate the relationship between ALAN exposure and habitat suitability for Arctiinae moths across the Neotropical region. We analyzed 17,989 occurrence records representing 3447 Arctiinae species, integrating bioclimatic variables and nighttime light data. We quantified the spatial overlap between potential species distributions and ALAN, including light intensity within occupied pixels. All species showed some level of exposure to artificial light, with 24% having more than 50% of their potential range affected. Species with broader geographic ranges exhibited disproportionately higher exposure. Predicted species richness was generally higher in illuminated areas, except in the Pantanal, indicating that regions of high ecological suitability often coincide with urbanized and artificially lit environments. The risks posed by ALAN to nocturnal biodiversity highlight the need for conservation strategies focused on areas with low levels of light pollution. Future studies integrating spatial, functional, and behavioral data will be essential to better understand both direct and indirect impacts of ALAN and to prioritize the conservation of nocturnal species in the Neotropical region.

**Keywords** Conservation · Insect decline · Model species distribution · Artificial light at night · Neotropical region · Vulnerability

## Introduction

The accelerated decline of insects, organisms essential to ecological processes such as pollination, nutrient cycling, and the maintenance of food webs, is driven by multiple anthropogenic pressures (Gebremariam 2024; Grevé et al. 2024). These include habitat loss and fragmentation resulting from agricultural and urban expansion, as well as pesticide use (Scolozzi and Geneletti 2012). Together with climate change, these pressures reduce both the abundance and diversity of organisms (Wagner 2022), undermining ecosystem stability and limiting the capacity of ecosystems to adapt to future disturbances (Dainese et al. 2019).

Understanding the impacts of anthropogenic pressures across taxonomic groups requires evaluating species vulnerability, defined as the interaction between exposure, sensitivity, and adaptive capacity to specific threats (De Marco et al. 2018). This framework can be treated as a

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spatial problem, in which exposure corresponds to the overlap between threats and species distributions (Dawson et al. 2011; Foden et al. 2013). Within this context, artificial light at night (ALAN) represents an important anthropogenic stressor currently undergoing rapid global expansion. More than 80% of the global population lives under light-polluted skies, with artificial sky brightness extending beyond urban areas (Falchi et al. 2016), while evidence indicates a continuous increase in its extent and intensity driven by urbanization and technological changes (Davies and Smyth 2018). In addition, ALAN reduces the visibility of natural night skies (Kyba et al. 2023) and can mask biologically relevant light cycles, such as lunar and twilight cues (Seymour et al. 2025). Together, these processes make ALAN a global driver of environmental change that increases species exposure within the vulnerability framework (De Marco et al. 2018).

Among the factors increasing insect vulnerability, artificial light at night (ALAN) has emerged as a major anthropogenic stressor. Emitted by sources such as streetlights, vehicle headlights, and urban infrastructure, ALAN disrupts ecosystems by altering the behavior and physiology of nocturnal and crepuscular species, thereby increasing their sensitivity to environmental change (Bará and Falchi 2023; Gaston et al. 2014). However, growing evidence suggests that ALAN can also affect diurnal species, either indirectly through changes in ecological interactions and resource availability or directly by disrupting circadian rhythms and behavior (Davies and Smyth 2018; Gaston and Holt 2018). In combination with land-use change, particularly the conversion of natural habitats into urban and agricultural landscapes (Falchi et al. 2016; Haddad et al. 2015), ALAN exacerbates habitat loss and fragmentation, with documented negative consequences for insect populations (Camacho et al. 2021; Owens et al. 2020).

ALAN has expanded rapidly over recent decades and now affects a substantial portion of the Earth's surface. Global assessments indicate that more than 80% of the world's population lives under light-polluted skies, with artificial sky brightness extending far beyond urban centers (Falchi et al. 2016). The increasing extent and intensity of nighttime illumination have led ALAN to be recognized as an important component of global environmental change, with consequences for ecological processes operating across multiple spatial scales (Davies and Smyth 2018). Recent evidence further suggests that artificial lighting continues to increase worldwide, contributing to a rapid decline in the visibility of stars and the natural darkness of night skies (Kyba 2023). Beyond increasing brightness, ALAN can mask biologically important natural light cycles, including lunar and twilight cues used by many organisms for orientation, movement,

reproduction, and other ecological functions (Seymour et al. 2025).

Moths are especially sensitive to ALAN and have consequently become model organisms for studying the ecological effects of light pollution (Boyes et al. 2020; Degen et al. 2024; Kawahara et al. 2018). This sensitivity is largely driven by their nocturnal habits and strong reliance on natural light cues for orientation and navigation. Artificial light profoundly influences moth behavior, often attracting individuals and promoting species aggregation in illuminated areas. Numerous hypotheses have been proposed to explain this flight-to-light behavior (Callahan 1965; Hamdorf and Höglund 1981; Mazokhin-Porshnyakov 1961; Robinson and Robinson 1950; Sothibandhu and Baker 1979). More recently, experimental evidence has shown that many flying insects exhibit a dorsal light response, in which they orient their dorsal side toward the brightest light source to maintain stable flight. Under artificial lighting, this mechanism can lead to disorientation, circling behavior, and prolonged attraction around light sources, providing a mechanistic explanation for the widespread aggregation of nocturnal insects around artificial lights (Fabian et al. 2024).

However, attraction to artificial light varies according to several factors, particularly the wavelength and type of lighting. Short-wavelength light, especially blue and ultraviolet spectra (370–400 nm) commonly emitted by cool white LEDs, causes strong sensory disruption, leading to disorientation, increased collision risk, and reduced survival (Cowan and Gries 2009; Donners et al. 2018; Frank 2006). Recent evidence suggests that these effects are driven by wavelength-specific physiological and behavioral responses, with Fabian et al. (2024) demonstrating that blue-rich light exacerbates navigational failure and energetic costs in moths. In contrast, longer-wavelength lighting, such as amber, red, or sodium-vapor lamps, exerts comparatively weaker effects on moth attraction and behavior (Fabušová et al. 2024).

Light responsiveness is further shaped by species-specific traits (Merckx and Slade 2014), as well as by body size and sex (Somers-Yeates et al. 2013; Van Langevelde et al. 2011). Larger, visually adapted individuals and males generally exhibit stronger attraction to artificial light (Garris and Snyder 2010). For males in particular, it remains unclear whether this pattern reflects heightened photoreceptive sensitivity or increased nocturnal activity and encounter rates with light sources (Garris and Snyder 2010). Owing to this strong phototactic response, light traps have become a widely used and highly effective method for surveying moth assemblages across multiple families (Justice et al. 2016; McDermott et al. 2018; Van Grunsven et al. 2014; Wölfling et al. 2016).

Beyond attraction alone, moth susceptibility to ALAN extends to disruptions of key physiological processes, such as diapause, even under low levels of nighttime illumination (Degen et al. 2024; Van Geffen et al. 2014). ALAN also alters behavioral and ecological dynamics, including reproduction and foraging efficiency, ultimately contributing to population declines and shifts in species distributions (Gaston and Holt 2018; Owens et al. 2020; Sanders et al. 2021; Van Grunsven et al. 2020).

This study focuses on Arctiinae moths (Lepidoptera: Erebiidae), a diverse subfamily comprising approximately 11,000 species worldwide (Zenker et al. 2016), of which around 6000 occur in the Neotropical region and 1650 in Brazil (Ferro and Diniz 2010). Arctiinae are notable for their chemically mediated defenses (e.g., sequestration of pyrrolizidine alkaloids), aposematism, mimicry, pronounced sexual dimorphism (Conner and Weller 2004; Weller et al. 1999), and their ability to emit ultrasonic signals as a defense against bat predation (Barber et al. 2009). They also represent one of the most widely used lepidopteran groups in environmental bioindication (Hilty and Merenlender 2000). Despite increasing research on the physiological and behavioral effects of light pollution, large-scale evaluations of the exposure of Arctiinae moths to artificial light at night (ALAN) are still limited. To address this gap, we quantified the degree of exposure of Arctiinae moths to ALAN across the Neotropical region by evaluating the overlap between species potential distributions and illuminated areas.

## Materials and methods

### Species occurrence records

Occurrence records of Arctiinae moths across the Neotropical region were obtained from online databases, including the Global Biodiversity Information Facility (GBIF; <https://www.gbif.org>) and speciesLink (<http://splink.cria.org.br/>), as well as from specimens collected by the authors and data compiled from the scientific literature (Delfina and Teston 2013; Ferro and Romanowski 2012; Grados et al. 2018; Moraes and Duarte 2015; Moreno and Ferro 2016; Nascimento et al. 2016; Silva and Silva 2014; Teston and Ferro 2016a, b, 2019, 2020).

In addition, we examined specimens deposited in Brazilian scientific collections, including the Coleção Becker (Camacan, BA), Coleção Entomológica do Instituto Nacional de Pesquisas da Amazônia, Coleção Entomológica Padre Jesus Santiago Moure (Universidade Federal do Paraná), Coleção Entomológica da Universidade de Brasília, Coleção do Laboratório de Estudos de Lepidópteros Neotropicais (Universidade Federal do Oeste do Pará),

Coleção Zoológica do Maranhão, Instituto Oswaldo Cruz (FIOCRUZ), Museu Entomológico Ceslau Biezanko (Universidade Federal de Pelotas), Museu de História Natural (Universidade Estadual de Campinas), Museu Nacional (Universidade Federal do Rio de Janeiro), Museu Paraense Emílio Goeldi, and the Museu de Zoologia da Universidade de São Paulo.

A total of 17,989 occurrence records representing 3447 Arctiinae species were compiled, distributed across 601 geographic coordinates in 24 countries. Prior to spatial analyses, all scientific names were carefully reviewed to correct spelling errors and resolve synonymies, ensuring taxonomic consistency and valid species-level identifications across all data sources. Records were then subjected to spatial cleaning and standardization, including conversion of coordinates to decimal degrees using the WGS84 datum, removal of records located outside the Neotropical region or in marine environments, and elimination of duplicate coordinates for the same species. These procedures were implemented using the CoordinateCleaner package (version 3.0.1) in R (version 4.3.1; R Core Team 2023).

When geographic coordinates were not available, approximate coordinates for collection localities were obtained using Google Maps, following Pessoa da Silva and De Marco (2023). At the end of this process, the final dataset comprised 227 species with eight or more occurrence records (Bean et al. 2012; Pearson et al. 2007), distributed across 17 countries and totaling 364 unique geographic coordinates within the Neotropical region. This minimum occurrence threshold was adopted to ensure the robustness of species distribution models, as both theoretical and empirical evidence indicate that models calibrated with few records tend to exhibit higher uncertainty, lower reliability, and reduced predictive performance (Bean et al. 2012; Pearson et al. 2007; Stockwell and Peterson 2002).

### Environmental data

We used 19 historical bioclimatic variables related to temperature and precipitation (1970–2000) obtained from WorldClim version 2.1 (<http://www.worldclim.org>) (Fick and Hijmans 2017), at a spatial resolution of 5 arc-minutes, corresponding to grid cells of approximately  $9.24 \times 9.24$  km. This resolution reduces spatial autocorrelation among the original variables while minimizing the influence of fine-scale biotic interactions on species distribution estimates (Lobo et al. 2010).

Because collinearity and high dimensionality among environmental predictors can lead to unstable model performance and increased uncertainty (De Marco and Nóbrega 2018), we performed a principal component analysis (PCA) using the ENMTML package (version 1.0.3; Andrade et al.

2020). The first eight principal components were retained, explaining approximately 96.5% of the total variance in the environmental dataset (Supplementary Table S1). The major axes primarily represented gradients associated with temperature, precipitation, and climatic seasonality, thereby summarizing the main environmental variation across the Neotropical region while reducing collinearity among predictor variables. (Supplementary Table S1). These components were subsequently used to calibrate ensemble species distribution models, resulting in continuous maps of environmental suitability for each species. Suitability values were then extracted and used as response variables in subsequent spatial analyses evaluating the overlap between species' potential distributions and artificial light at night.

### Light pollution data

Light pollution data were obtained from the NOAA Defense Meteorological Satellite Program–Operational Linescan System (DMSP-OLS) Nighttime Lights dataset, available through the Google Earth Engine (GEE) platform. This dataset provides records of nocturnal light emissions captured by the *Operational Linescan System* (OLS) sensor aboard *Defense Meteorological Satellite Program* (DMSP) satellites. Each DMSP-OLS image corresponds to a cloud-free annual composite of nighttime observations acquired during hours of darkness. The data have a nominal spatial resolution of approximately 1 km at the equator, resampled from an original footprint of ~2.7 km, and pixel values are expressed as digital numbers (DN) ranging from 0 to 63, which serve as comparative indicators of light intensity. These digital number (DN) values represent relative radiance and are widely used as proxies for artificial light intensity in large-scale ecological studies. A value of 0 indicates the absence of detectable light, whereas higher values represent increasing levels of persistent artificial illumination; highly illuminated urban areas typically saturate at DN = 63 (Bennie et al. 2014).

Data extraction and processing were conducted using the GEE programming interface integrated with R via the *rgee* package (Aybar et al. 2020), which enables direct access to and manipulation of geospatial datasets hosted on the platform. The image collection was filtered for the year 2013, the most recent year available for DMSP-OLS data, and the band corresponding to stable lights (*stable\_lights*) was selected, representing persistent light sources such as urban public lighting. We acknowledge that the use of data from a single year does not capture temporal variation in light intensity or spatial distribution, and that changes may have occurred before or after 2013, potentially affecting species records collected over multiple years. Nevertheless, DMSP-OLS Stable lights constitute a widely used proxy for

persistent anthropogenic nighttime illumination and urban development patterns at regional and continental scales (Small et al. 2011; Li and Zhou 2017). Therefore, despite this limitation, the 2013 dataset is suitable for assessing general spatial associations between artificial light at night (ALAN) and species occurrence.

We processed the nighttime light data (in GeoTIFF format) in R using the *terra* package version 1.8 (Hijmans 2021) (Supplementary Fig. S1). Initially, we reprojected the shapefile of the Neotropical region to match the coordinate reference system (CRS) of the raster and subsequently rasterized using the *rasterize* function, ensuring compatibility with the species distribution models. For spatial standardization, the data were adjusted to the cell resolution of the models (~9.24 km × 9.24 km) and to the same geographic extent, ensuring precise overlap. Finally, we exported the processed rasters again as GeoTIFFs, preserving geospatial metadata for subsequent analyses. For this specific analysis, light pollution data were binarized to distinguish the presence versus absence of artificial illumination, with pixels > 0 considered illuminated. This approach was adopted to assess broad-scale spatial overlap between species' potential distributions and permanent artificial light and does not account for variation in light intensity.

### Potential species distribution models

To reduce uncertainty and increase the robustness of species distribution model results, we used four algorithms for modeling: *Bayesian Gaussian* – GAU (Golding 2014), *Maximum Entropy* – MaxEnt (Phillips et al. 2017), *Random Forest* – RDF (Liaw and Wiener 2002), and *Support Vectorial Machine* – SVM (Karatzoglou et al. 2004). These algorithms encompass different theoretical bases for estimating habitat suitability and are widely used in species distribution modeling (Beeman et al. 2021; Sillero et al. 2021; Velazco et al. 2019). According to Barbet-Massin et al. (2012), the way pseudo-absences are distributed across geographic space can influence model performance. In this study, pseudo-absences were generated using an environmental constraint method to allocate them in regions distinct from the environmental space occupied by the species. This method helps to reduce bias in model training by avoiding areas that are likely suitable but unsampled. The pseudo-absence data were applied to all the algorithms mentioned above.

We used two data partitioning protocols. For species with fewer records (between 8 and 20), models were fitted using bootstrap. The dataset was split into training (70% of the records) and testing (30% of the records) subsets. For species with more than 20 records, the checkerboard method was applied. In this method, geographic space is divided into blocks, and occurrence data are split into two

subgroups for model fitting and evaluation (Andrade et al. 2020). Sørensen's similarity index (Li and Guo 2013; Sørensen, 1948) was used as a measure of model reliability. This index ranges from 0 to 1, with values closer to 1 indicating more accurate models (Araújo et al. 2018). For each species, the *ensemble* model was calculated as the mean suitability of models performing at or above the mean Sørensen index value for each algorithm (Pessoa da Silva and De Marco 2023).

To reduce omission (underprediction) and commission (overprediction) errors, the M-SDM method was applied a posteriori using the PRES technique, which restricts predictions to locations supported by occurrence records. This approach reduces the inclusion of environmentally suitable areas that may not be occupied due to historical, biogeographic, or dispersal constraints, thereby providing a more realistic representation of species distributions (Mendes et al. 2020). All analyses related to species distribution modeling were conducted using the ENMTML package version 1.0.3 (Andrade et al. 2020) in R 4.3.1 (R Core Team 2023).

## Analytical procedures

To investigate the degree of overlap between the potential distribution of each species and light pollution, we conducted a spatial analysis in R. Initially, the raster files of species' potential distributions were imported from a folder containing tiff files (Pessoa da Silva and De Marco 2023). The light pollution layer was resampled to match the same extent (the Neotropical region) and spatial resolution (cells of  $9.24 \times 9.24$  km) as the generated models. For this specific analysis, light pollution data were binarized to distinguish the presence versus absence of artificial illumination, with pixels  $> 0$  considered illuminated. This approach was used to assess broad-scale spatial overlap and does not account for variation in light intensity. For each species, the spatial overlap between its potential distribution and illuminated areas was assessed.

For the primary analysis, light pollution data were binarized to distinguish illuminated from non-illuminated areas, with pixels  $> 0$  classified as exposed to artificial light. This binary approach was used to quantify broad-scale spatial overlap between species' potential distributions and ALAN, independent of variation in light intensity. In addition, we characterized species' exposure to ALAN using continuous intensity values, including mean, minimum, and maximum light levels across each species' distribution. The total number of occupied pixels (as a proxy for potential distribution area), as well as the proportion of illuminated pixels within each species' range, were also calculated to describe spatial exposure patterns. To test whether species with larger potential distributions experience higher exposure to artificial

light, we fitted a linear regression model relating distribution size (number of occupied pixels) to mean light intensity across species distributions, treating each species as an independent replicate.

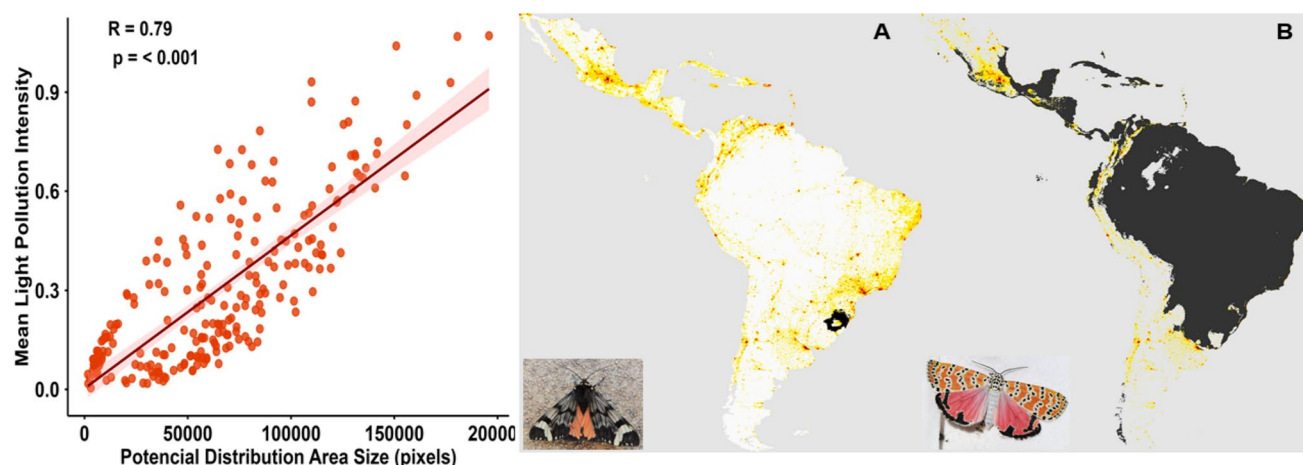
To assess the potential influence of light pollution on biodiversity patterns, the exposure of species richness to artificial light at night (ALAN) was estimated for each Brazilian biome (Amazon, Atlantic Forest, Cerrado, Caatinga, Pantanal, and Pampa), as defined by the official classification and spatial boundaries provided by the Brazilian Institute of Geography and Statistics (IBGE), and for the entire Neotropical region. This exposure was calculated as the difference between the mean species richness in illuminated and non-illuminated pixels. To quantify the uncertainty associated with this difference, 95% confidence intervals CI were estimated for light exposure. The lower and upper bounds were calculated by propagating the error from the CI already obtained for the illuminated and non-illuminated groups.

## Results

All evaluated moth species showed some level of overlap between their potential distribution area and regions with artificial light. The percentage of overlap ranged from 0.25% to nearly 89%. Fifty-four species (24%) had more than 50% of their predicted distribution area within artificially illuminated areas. *Utetheisa ornatrix* (Linnaeus 1758) exhibited the highest overlap rate (88.7%), followed by *Idalus carinosa* (Schaus 1905) and *Aclytia heber* (Cramer 1780) (both 80%). Approximately 18% of the species showed less than 10% overlap between illuminated and non-illuminated areas. Among these, *Dysschema hilara* (Weymer 1895) exhibited the lowest overlap, with only 0.25% (Supplementary Table 2). A positive and significant relationship was observed between light pollution intensity and species' distribution area (Fig. 1), indicating that more widely distributed species are proportionally more exposed to areas with high levels of artificial brightness.

We analyzed the spatial overlap between species occurrence points and the light pollution map in the study region. The highest light pollution values are primarily concentrated in urban areas and densely populated regions, whereas peripheral and less urbanized areas exhibit low levels of artificial illumination. Species occurrences are distributed heterogeneously with respect to light pollution. A significant portion of occurrences is located in areas with low to moderate light pollution. However, some occurrences are also found in areas with high levels of artificial light (Supplementary Figure S2).

The predicted mean richness of Arctiinae moths in illuminated areas was higher than in non-illuminated areas



**Fig. 1** Relationship between light pollution intensity and the size of the potential distribution area of 227 Arctiinae moth species in the Neotropical region. Each circle represents a species; the red line indicates the simple linear regression, and the shaded area represents the 95% confidence interval. **A** Potential distribution maps (in gray) of the species with the smallest range (*Dysschema hilara*) overlaid on the light pollution intensity map (yellow to red shades); **B** Potential distribution maps (in gray) of the species with the largest range (*Utetheisa ornatrix*) overlaid on the light pollution intensity map (yellow to red shades). Species: *Dysschema hilara* Photo by amandapmarcon/ iNatu-

alist, licensed under CC BY-NC. Species: *Utetheisa ornatrix*. Photo by Zachary Lang / iNaturalist, licensed under CC BY-NC. We emphasize that ALAN intensity exhibits a strongly right-skewed distribution across the landscape, with most pixels showing very low values and a small proportion of highly illuminated urban pixels. Therefore, the mean ALAN values presented here are used as an exploratory metric for comparability among species and should not be interpreted as a complete representation of typical exposure. Complementary metrics, including median values and the proportion of illuminated pixels within species' ranges, were also calculated (see Methods)

across the Neotropical region and in five Brazilian biomes (Fig. 2A). However, the opposite pattern was observed in the Pantanal biome (Fig. 2A). Overall, the difference between mean richness values in illuminated and non-illuminated areas was low (between 5 and 10%) (Fig. 2B). In the Caatinga, however, there were on average 30% more species in areas with light than in areas without light (Fig. 2B).

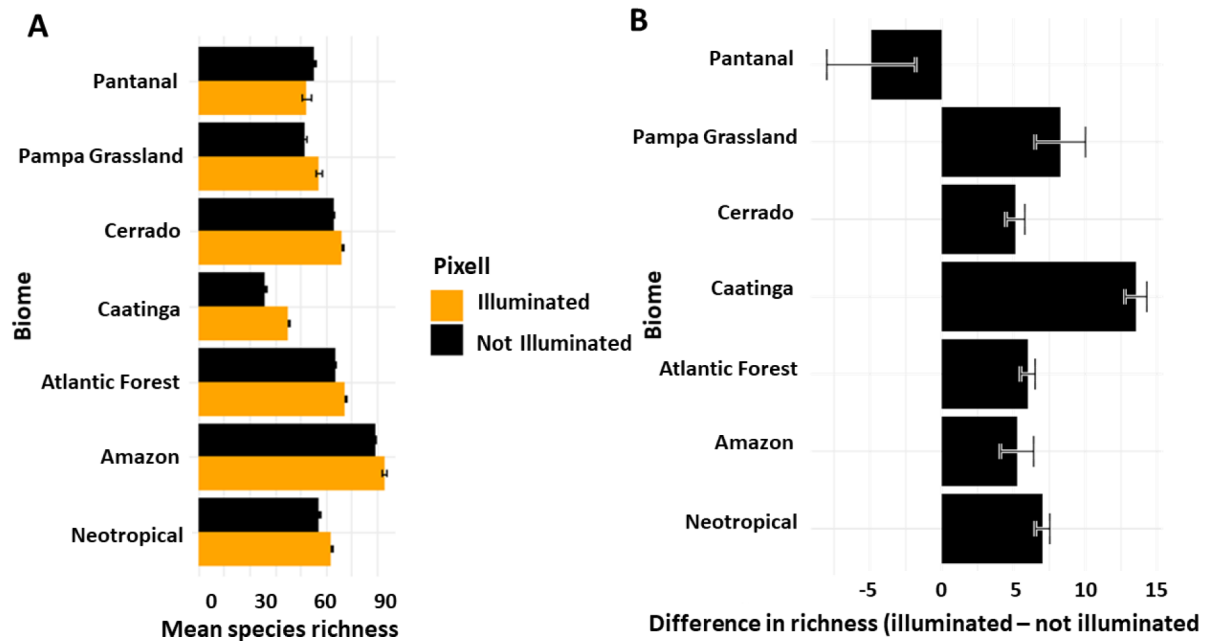
The highest estimates of Arctiinae richness are primarily concentrated in the Amazon region (Fig. 2A and 3). In contrast, higher-latitude regions, such as southern Brazil, Uruguay, Argentina, Chile, and the Andean region, exhibit the lowest richness values (Fig. 3). Additionally, pixels associated with light pollution are notably concentrated in coastal areas (e.g., the Brazilian Atlantic Forest) and major urban centers (such as the capitals of Mexico, Venezuela, and Argentina) (Fig. 3). Overall, spatial patterns suggest that light pollution partially overlaps with areas of high richness, particularly in biomes such as the Atlantic Forest, whereas in regions of extremely high richness, such as central Amazonia, the overlap is less intense.

## Discussion

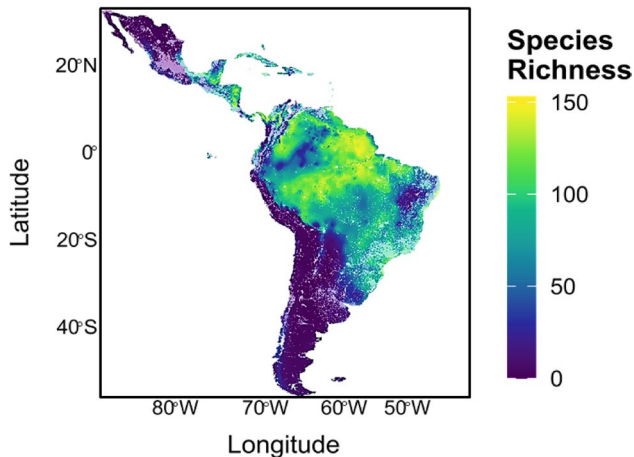
Although the effects of Artificial Light at Night (ALAN) on nocturnal fauna have been receiving increasing attention (Boyes et al. 2021; Fabian et al. 2024; Fabusova et al. 2024; Hakbong et al. 2021; Kalinkat et al. 2021a, b; MacGregor et al. 2015; MacGregor et al. 2017; Owens et al.

2018; Owens et al. 2024; Seymoure et al. 2025; Van Koppenhagen et al. 2025a; Van Langevelde et al. 2017), our study is the first to integrate large-scale species distribution modeling with ALAN exposure. All Arctiinae species evaluated here exhibit some exposure to artificial light at night within their potential distribution areas, highlighting the growing concern over the impacts of light pollution on nocturnal biodiversity (Boyes et al. 2020; Fabusova et al. 2024; MacGregor et al. 2017; Owens et al. 2018). Although DMSP-OLS digital number (DN) values (0–63) represent relative radiance rather than direct measurements of illuminance, they can be interpreted in the context of natural nocturnal light. Low DN values correspond to conditions close to natural darkness (e.g., new moon), intermediate values to moonlit conditions, and high values to levels of artificial illumination that exceed natural night-time light by several orders of magnitude. This interpretation is consistent with studies that compare artificial light to natural light sources or standardized sky brightness metrics (e.g., Falchi et al. 2016; Seymoure et al. 2025), allowing ALAN to be interpreted as a gradient of replacement of natural night-time darkness conditions by artificial illumination.

Species with broader distributions were proportionally more exposed to ALAN, likely because their larger geographic ranges increase the probability of overlapping with urban centers, transportation corridors, and other human-modified landscapes where artificial lighting is concentrated. This pattern indicates that exposure to artificial light is not restricted to narrowly distributed species and that even



**Fig. 2** **A** Mean potential species richness of Arctiinae moths in illuminated and non-illuminated pixels across six Brazilian biomes and the entire Neotropical region. **B** Difference in mean species richness between illuminated and non-illuminated pixels (illuminated–non-illuminated). Positive values indicate higher richness in illuminated areas, whereas negative values indicate higher richness in non-illuminated areas. Error bars represent 95% confidence intervals



**Fig. 3** Potential species richness of Arctiinae moths in the Neotropical region. Gray areas indicate pixels with artificial light at night (ALAN) pollution

widely distributed taxa may experience substantial exposure across portions of their range. As urban areas continue to expand and artificial lighting becomes increasingly pervasive, ALAN may represent a growing pressure on nocturnal biodiversity at continental scales, affecting species across a broad spectrum of geographic distributions (Li et al. 2022; Ren et al. 2023; Seto et al. 2012).

The vulnerability of a species to light pollution is assessed based on its degree of exposure, sensitivity, and

capacity for adaptation or persistence under artificial light. In our set of moths, we found that the degree of exposure is high (exceeding 50% of the potential distribution area) for 54 species. These species belong to various subtribes and exhibit different levels of dietary specialization, body sizes, types of chemical defense, and activity periods. However, due to the absence of studies investigating the sensitivity or adaptive capacity of these species to light, it is not yet possible to quantify their overall vulnerability to exposure to ALAN.

Recent evidence indicates that larger moths are more strongly affected by light pollution, exhibiting steeper declines in naturally dark regions than in already illuminated urban areas (Know et al. 2024). This pattern suggests that species adapted to naturally dark environments may be particularly sensitive to the introduction of artificial light, even at relatively low levels, because it represents a substantial alteration of natural nighttime conditions. Such changes can interfere with orientation, reproductive behavior, and increase predation risk (Boyes et al. 2021; Slade et al. 2013). Consequently, although ALAN may affect a broad range of species, its impacts are likely to be disproportionately greater for more sensitive groups within the Arctiinae.

Alarmingly, estimates of Arctiinae richness were 5 to 35% higher in areas exposed to ALAN compared to non-illuminated areas. This pattern remained consistent across

the entire Neotropical region and across most Brazilian biomes evaluated (Caatinga, Pampa, Atlantic Forest, Cerrado, and Amazon). However, this result should not be interpreted as evidence that artificial lighting benefits moth diversity. Instead, it indicates that a substantial proportion of the potential distribution of Arctiinae overlaps with illuminated areas, exposing many species to a well-documented anthropogenic stressor.

Because species richness estimates were derived from potential distributions generated by species distribution models (SDMs), which are based primarily on climatic suitability, rather than from observed population responses, the higher richness found in illuminated areas reflects a greater spatial overlap between biodiversity-rich regions and areas exposed to artificial light, rather than a positive ecological effect of ALAN. Consequently, the realized richness of Arctiinae in these areas may be substantially lower than predicted, as artificial light at night and other disturbances associated with urbanization may reduce local population persistence or prevent species from occupying habitats that are otherwise climatically suitable. Therefore, the overlap between high predicted species richness and illuminated areas should be interpreted as an indication of potential conservation risk, rather than evidence that these areas maintain equally high levels of realized biodiversity.

Since light pollution is typically associated with urbanization (Henderson et al. 2009; Elvidge et al. 2013; Bickenbach et al. 2014), species are likely to be more exposed not only to artificial light but also to various other disturbances characteristic of urban environments, such as chemical pollution (e.g., CO<sub>2</sub>, insecticides), noise pollution (Arctiinae detect and emit ultrasonic signals), reduced food availability, and microclimatic changes.

Beyond light intensity and spatial extent, the ecological consequences of ALAN also depend strongly on its spectral composition and lighting characteristics. Different lighting technologies emit distinct wavelength distributions, resulting in varying levels of attraction and behavioral disruption in nocturnal insects. Short-wavelength-rich (blue-enriched) light has been reported to elicit stronger attraction in moths than longer-wavelength light, and may also produce stronger behavioral and physiological effects in some nocturnal insects (Van Langevelde et al. 2011; Somers-Yeates et al. 2013). However, responses to light colour are not necessarily consistent across species or experimental contexts, and some studies have found limited or contrasting effects of wavelength on moth activity and other ecological responses. Thus, although spectral composition is an important component of ALAN, its effects may depend on species-specific sensory and behavioural characteristics, as well as on other properties of the light source. Likewise, lighting design, including shielding, beam directionality, and light spill,

determines how far artificial light penetrates surrounding natural habitats, thereby influencing the spatial extent of ecological disturbance (Owens et al. 2024; Van Koppenhagen et al. 2025a). Consequently, ALAN can disrupt orientation, movement, pheromone-mediated mate attraction, and ultrasonic communication, reducing reproductive success and altering ecological interactions such as plant–pollinator networks and other trophic processes that support moth populations (Altermatt et al. 2009; Boyes et al. 2020; MacGregor et al. 2017; Owens et al. 2018; Van Geffen et al. 2015a, b; Van Langevelde et al. 2017).

In the Pantanal, species richness was higher in areas without ALAN than in illuminated areas, suggesting that dark areas may play an important role in maintaining nocturnal biodiversity. Although the mechanisms underlying this pattern remain uncertain, the introduction of artificial lighting into landscapes that historically experienced low levels of nighttime illumination may disrupt the behavior and ecological interactions of nocturnal insects. The predominantly open vegetation of the Pantanal may allow artificial light to spread over greater distances than in densely forested environments, potentially increasing the spatial extent of ALAN exposure (Kyba et al. 2015; Jechow et al. 2021). The Pantanal is also experiencing increasing environmental pressures, including prolonged droughts, more frequent and intense wildfires, and extensive land-use change, with approximately 45% of its original area already modified by human activities (Alho et al. 2011; Tomas et al. 2019).

The Amazon region, southern Mexico (Yucatán Peninsula), and southern Central America (Panama, Nicaragua, and Honduras) harbor the highest potential richness of Arctiinae species. Coincidentally, these regions currently exhibit low levels of exposure to ALAN. However, urbanization in the Amazon has been rapidly advancing, transforming ecosystems through the expansion of roads, agriculture, and mining activities (Fenzl et al. 2020), which may accelerate the spread of ALAN into previously isolated areas. Therefore, as suggested for the Pantanal, the establishment of new protected areas and the preservation of existing ones would be highly recommended.

This study opens new perspectives by highlighting the impacts of ALAN on nocturnal biodiversity in the Neotropical region, emphasizing the vulnerability of Arctiinae moths. The overlap between areas of high species richness and artificially illuminated zones indicates significant ecological risks, which are likely to intensify with urban expansion and large-scale development projects in the Amazon and other biomes. Future research should expand monitoring efforts in under-sampled regions and integrate spatial, functional, and behavioral data to better understand the responses of moths and other nocturnal organisms to the direct and indirect effects of ALAN. In addition, an important limitation

of current large-scale assessments is that satellite-based datasets primarily quantify the intensity and spatial extent of nighttime illumination but generally cannot distinguish the type of lighting technology or its spectral composition. This limitation is particularly relevant because different light spectra and lighting technologies elicit distinct behavioral and physiological responses in nocturnal insects. Consequently, areas with similar light intensities may differ substantially in their ecological impacts. Future satellite products and large-scale analyses integrating spectral information will therefore be essential for improving our understanding of the mechanisms through which ALAN affects biodiversity and for developing more effective conservation and lighting management strategies.

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**Data availability** The data used in this study are publicly available, as indicated in the Materials and Methods section. Species occurrence records used to develop the distribution models were obtained from publicly accessible databases (Global Biodiversity Facility - [<https://www.gbif.org/>] (<https://www.gbif.org>) and speciesLink - [<http://splink.cria.org.br/>] (<http://splink.cria.org.br>)).

## Declarations

**Conflict of interest** The authors declare no competing interests.

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